

Associations of proton pump inhibitors with susceptibility to influenza, pneumonia, and COVID-19: Evidence from a large population based cohort study

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Abstract

Adverse effects of proton pump inhibitors (PPIs) have raised wide concerns. The association of PPIs with influenza is unexplored, while that with pneumonia or COVID-19 remains controversial. Our study aims to evaluate whether PPI use increases the risks of these respiratory infections. The current study included 160,923 eligible participants at baseline who completed questionnaires on medication use, which included PPI or histamine-2 receptor antagonist (H2RA), from the UK Biobank. Cox proportional hazards regression and propensity score-matching analyses were used to estimate the hazard ratios (HRs) and 95% confidence intervals (CIs). Comparisons with H2RA users were tested. PPI use was associated with increased risks of developing influenza (HR 1.32, 95%CI 1.12-1.56) and pneumonia (hazard ratio [HR] 1.42, 95% confidence interval [CI] 1.26-1.59). In contrast, the risk of COVID-19 infection was not significant with regular PPI use (HR 1.08, 95%CI 0.99-1.17), while the risks of severe COVID-19 (HR 1.19, 95%CI 1.11-1.27) and mortality (HR 1.37, 95%CI 1.29-1.46) were increased. However, when compared with H2RA users, PPI users were associated with a higher risk of influenza (HR 1.74, 95%CI 1.19-2.54), but the risks with pneumonia or COVID-19-related outcomes were not evident. In conclusion, PPI users are associated with increased risks of influenza, pneumonia, as well as COVID-19 severity and mortality compared to non-users, while the effects on pneumonia or COVID-19-related outcomes under PPI use were attenuated when compared to the use of H2RAs. Appropriate use of PPIs based on comprehensive evaluation is required.

eLife assessment

This **useful** study aimed to quantify associations between regular use of proton-pump inhibitors (PPI) with the occurrence of respiratory infections, such as influenza, pneumonia, COVID-19, and others over a period of several years. PPI use was associated with increased risks of influenza, pneumonia, but not of COVID-19, although severity and mortality of COVID-19 infections were higher in PPI users. There are inevitable weaknesses of the study design used, such as the fact that PPI use was only measured at one time-point whereas infections were assessed over a long time period, but these are appropriately highlighted in the discussion. Weaknesses are highlighted in the discussion and the study presents **convincing** evidence for the conclusions overall.

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1. Introduction

Proton pump inhibitors (PPIs), as one of the most commonly used medications worldwide, have been utilized for treating various conditions related to excessive gastric acid secretion [1]. In the United States, the prescription of PPIs has doubled from 1999 to 2012, and the number of people taking PPIs is still increasing due to their availability over the counter [2]. However, concerns are being raised regarding the long-term and inappropriate use of PPIs, which have been linked to a wide range of adverse conditions, including osteoporotic fractures, renal failure, and vitamin deficiencies [3–5].

PPI-induced hypochlorhydria and gastrointestinal residence of pathogens might increase the risk of respiratory infections [6]. Cohort studies in the United Kingdom and the United States reveal the risks of developing community- and hospital-acquired pneumonia are increased by approximately 100% and 30%, respectively [7, 8]. In contrast, a nested case-control study based on the UK General Practice Research Database indicates long-term PPI therapy is not associated with increased risk for community-acquired pneumonia [9], and a retrospective cohort study involving 593 265 patients in Canada demonstrates no increased risk in developing pneumonia among PPI recipients [10]. Recently, attention has also been paid to the susceptibility to Coronavirus Disease 2019 (COVID-19) caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Based on 53,130 participants in the United States, a dose-dependent increased risk of COVID-19 positivity among PPI users is found [11]. Another Danish study also indicate a marginally increased risk of COVID-19 positivity [12], whereas studies based on the UK Biobank and a Korean cohort indicate nonsignificant association [13, 14]. Meta-analyses on the associations between the use of PPI and SARS-CoV-2 infection also demonstrate conflicting results [12, 15].

To date, the association between PPI and influenza remains unknown. The current evidence referring to PPI, pneumonia and COVID-19 is controversial. Previous studies had several limitations. For instance, the study based on the UK General Practice Research Database did not adjust for potential confounding variables for PPI indications [16]. Direct comparisons with histamine-2 receptor antagonist (H2RA) users were not conducted in previous studies to minimize confounding by indication. In addition, the findings by Almario *et al.* were based on patients with gastrointestinal symptoms, rather than the general population [11]. The previous UK Biobank study merely included participants tested for COVID-19 from March to June 2020 [13].

By leveraging the large-scale cohort and updated data in the UK Biobank, we aim to evaluate the associations between the regular use of PPIs and the susceptibility to respiratory infections, including influenza, pneumonia, and COVID-19.

2. Methods

2.1 Study population

The detailed information on study design for the UK Biobank was described previously [17]. Invitations were sent to about 9.2 million people who were aged 40-69, had capacity to consent, registered with the National Health Service (NHS), and lived within 25 miles of one of the assessment centers [18]. The participants were free to withdraw at any time [17]. Over 0.5 million participants were recruited from 22 assessment centers in Scotland, England, and Wales (specific locations of assessment centers are available at: <https://biobank.ndph.ox.ac.uk/ukb/field.cgi?id=54>) from 2006 to 2010. Information such as touch screen questionnaire, interview, blood pressure, eye measurements, physical measurements and so on was collected in the assessment centers (detailed content of assessments is available at: <https://biobank.ndph.ox.ac.uk/ukb/refer.cgi?id=100241>). Written informed consent was acquired from each participant, and ethical approval was obtained from the North West Multi-Center Research Ethics Committee (approval number: 11/NW/0382, 16/NW/0274, and 21/NW/0157). The current study has been approved under the UK Biobank project 83339. In this study, 11,171 participants with missing PPI medication data and 56,907 participants with missing covariate data were excluded, and we further restricted the cohort to the participants with available primary-care data. Among them, 1,297 participants without follow-up, which were mainly determined by reported death, departure from the UK, or withdrawn consent, had been removed after initial exclusion. For the evaluation of associations with influenza, pneumonia, and other respiratory infections, those with outcomes occurred before the baseline, or only with self-reported records and diagnoses were further excluded. For the COVID-19 infection and COVID-19-related outcomes, we excluded participants whose COVID-19 testing data were unavailable or who had died before the COVID-19 pandemic (**Figure 1**).

2.2 Definition of exposure

The exposure of interest was regular use of PPIs. The participants could enter the generic or trade name of the treatment on the touchscreen to match the medications they used (**Supplementary Table S1**). Verbal interviews were conducted by a trained nurse if participants answered that they were regularly taking prescribed or over-the-counter medication on the touchscreen, in which 'regular' was defined as most days of the week for the last 4 weeks, and information on specific types of medications was further recorded, while no response to the question on the interview was considered missing data for PPI use. Short-term medications, for example, a 1-week course of antibiotics, were not recorded in the interview. Types of PPIs available in the UK Biobank included omeprazole, lansoprazole, esomeprazole, rabeprazole, and pantoprazole. The regular use of H2RAs was also defined by the above process. When comparing PPI users with H2RA users, participants who took both medications were excluded. Information on dose or duration of acid suppressant use was not available in the UK Biobank.

2.3 Definition of outcomes

The primary outcomes of interest were influenza, pneumonia, COVID-19 infections (**Supplementary Table S2**). Briefly, the first reported occurrences of respiratory system-related conditions within primary care data and hospital inpatient data defined by the International Classification of Diseases (ICD)-10 codes were categorized by the UK Biobank (<https://biobank.ndph>

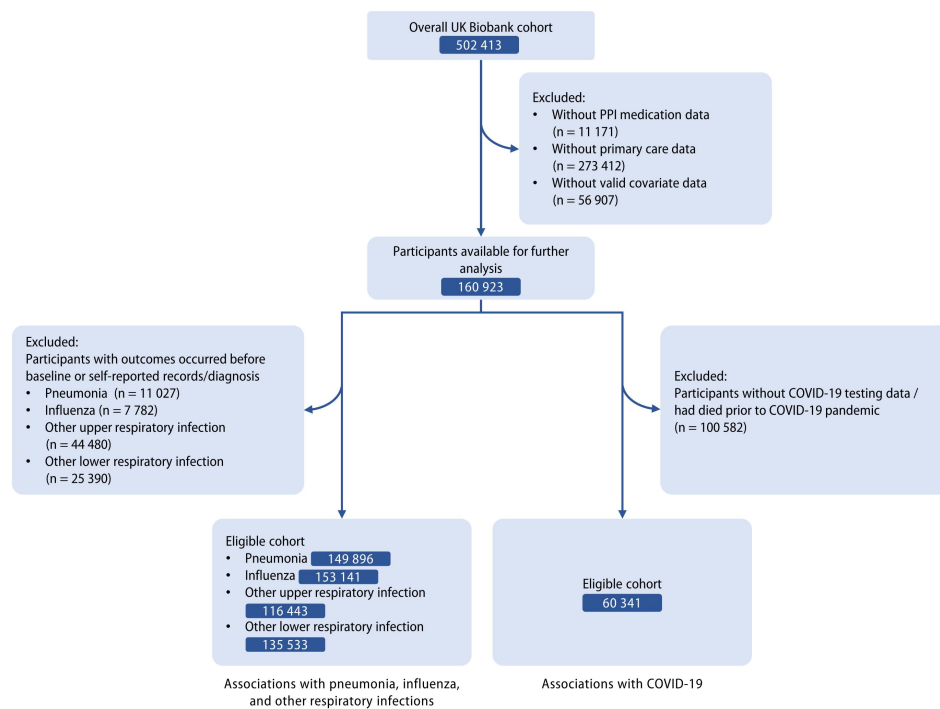


Figure 1.

Flow diagram of eligible participants selection.

[.ox.ac.uk/showcase/label.cgi?id=2410](https://www.ox.ac.uk/showcase/label.cgi?id=2410)). Influenza included those caused by identified influenza virus (J09-J10) and virus not identified (J11). Pneumonia was defined as that caused by viruses (J12), bacteria (J13-15), and other infectious organisms (J16-18).

COVID-19-related data in the UK Biobank (available from January 2020 to September 2021) based on follow-up of the participants was used [19]. COVID-19 infection mainly included information on positive COVID-19 tests, and patients with inpatient diagnoses or mortality due to COVID-19 were also regarded as having COVID-19 infection.

The secondary outcomes included other upper or lower respiratory infections, COVID-19 mortality, and COVID-19 severity. The definition of other upper respiratory infections contained acute nasopharyngitis, sinusitis, pharyngitis, tonsillitis, laryngitis, tracheitis, obstructive laryngitis, epiglottitis, or upper respiratory infections of multiple and unspecified sites (J00-J06). Other lower respiratory infections included acute bronchitis, bronchiolitis, and other unspecified ones (J20-J22). Severe COVID-19 cases were defined as being hospitalized for COVID-19. COVID-19 mortality was defined as the underlying recorded cause of death due to COVID-19 (ICD-10 U07.1 and U07.2).

2.4 Assessment of covariates

The covariates used for adjustments in our study were identified by a directed acyclic graph (DCA, **Supplementary Figure 1**) based on existing literature and expert knowledge. Baseline data on sociodemographic information (age, sex, ethnicity), socioeconomic status (deprivation index, which was defined using national census information on car ownership, household overcrowding, owner occupation, and unemployment combined for postcode areas of residence), alcohol consumption, smoking status, fresh fruit intake, multivitamin use, and body mass index (BMI) were collected from the UK Biobank, while physical activity was assessed using the International Physical Activity Questionnaire-Short Form. Gastroesophageal reflux disease (GERD), peptic ulcers, and upper gastrointestinal bleeding, were included as they are main indications for the use of PPIs. The comorbidities (hypertension, type 2 diabetes, renal failure, myocardial infarction, stroke, chronic obstructive pulmonary disease [COPD], asthma) were examined using self-reported data and adjusted due to their impact on the risk of respiratory infections. Since PPI and H2RA have highly similar indications, the use of H2RA was also adjusted. Data on medication use including aspirin, non-aspirin non-steroidal anti-inflammatory drugs (NSAIDs, including ibuprofen), and cholesterol-lowering medications were extracted and adjusted. For influenza and COVID-related outcomes, vaccinations were additionally adjusted. The linearity between continuous variables and outcomes was assessed by Martingale residuals plots, while the variables detected with non-linearity were regarded as categorical variables for further analyses.

2.5 CYP2C19 genetic variants

PPIs are mainly cleared by *CYP2C19*, and therefore their metabolism and effects are affected by different variants of *CYP2C19*. Genotyped genetic variant data after quality control was available for UK Biobank participants based on the Affymetrix Axiom UKB array and the Affymetrix UKBiLEVE array [20]. According to the Clinical Pharmacogenetics Implementation Consortium Guideline for *CYP2C19* and Proton Pump Inhibitor Dosing [21], genotypic data of four *CYP2C19* variants, including *rs12248560* (*CYP2C19**17), *rs17884712* (*CYP2C19**9), *rs4986893* (*CYP2C19**3), and *rs4244285* (*CYP2C19**2), were utilized to divide PPI users into three subgroups: (1) *CYP2C19* rapid and ultrarapid metabolizers (carried 1 functional allele and 1 increased-function allele [*17]; or carried 2 increased-function alleles); (2) *CYP2C19* normal metabolizers (carried 2 functional alleles); (3) *CYP2C19* likely intermediate, intermediate and poor metabolizers (carried ≥ 1 alleles with no/decreased function [*2, *3, and *9]).

2.6 Statistical analysis

The baseline characteristics were demonstrated by percentages for categorical variables, and mean (standard deviation [SD]), or median (interquartile range [IQR]) for continuous variables according to the distribution of data after evaluating the data distribution.

Univariate and multivariable Cox proportional hazards regression models were utilized to assess the association between regular use of PPIs and the selected outcomes, and the results were presented as hazard ratios (HRs) with 95% confidence intervals (CIs). Multivariable model 1 included age and sex. Model 2 additionally contained other potential confounders selected *a priori*, including ethnicity, deprivation index, alcohol consumption, smoking, physical activity, BMI, fresh fruit intake, GERD, peptic ulcer, upper gastrointestinal bleeding, hypertension, type 2 diabetes, renal failure, myocardial infarction, stroke, COPD, asthma, aspirin, non-aspirin NSAIDs, H2RA, cholesterol-lowering medications, and multivitamin use. The reference group was participants without regular use of PPIs.

Schoenfeld residuals tests were used to evaluate the proportional hazards assumptions, while no violation of the assumption was detected (**Supplementary Table S3**). Person-years were calculated from the number of participants and the date from Jan 2020 (for COVID-19-related outcomes) or recruitment (for other respiratory outcomes) to outcome diagnosis, last follow-up (September 2021 for COVID-19 infection and related outcomes; December 2021 for other outcomes), or death, whichever came first. Stratified analyses according to population characteristics, types of PPIs, and *CYP2C19* metabolizers were performed using multivariable-adjusted models across subgroups of each stratifying variable, and the multiplicative interactions were evaluated using likelihood ratio tests.

Quantitative bias analyses were performed to calculate e-values, which illustrates the strength of association between an unmeasured confounder and exposure or outcome, conditional on the measured covariates [22]. E-value is the smallest magnitude of risk estimates that an unmeasured confounder would need to have with the exposure and outcome to explain away an observed association [22]. The event-free probabilities were compared by Kaplan-Meier survival curves with inverse probability weights adjusting for the measured covariates. In addition, we conducted additional analyses using multiple imputation by chained equations to include participants initially excluded due to missing ethnicity data using the “mice” package [23]. Self-reported infections, except for COVID-19-related outcomes due to the lack of data, were also included for the outcomes as sensitivity analyses. The self-reported cases were reported at the baseline or subsequent UK Biobank assessment center visit. Moreover, propensity score-matching analysis was conducted. The same set of covariates was used to derive propensity scores, and the PPI users and non-users were matched with a ratio of 1:4 using the “MatchIT” package [24], which estimated the propensity scores in the background and matched observations based on the nearest neighbor method. The remaining imbalanced covariates (standardized mean difference ≥ 0.1) after propensity score matching were further adjusted by multivariable Cox regression models to calculate HRs and 95% CIs [25]. Furthermore, because PPI and H2RA share highly similar indications, we performed head-to-head comparisons between PPI and H2RA users to further minimize the protopathic bias [26, 27].

All statistical analyses were performed using R (version 4.2.0, <https://www.r-project.org/>). The significance level at $\alpha = 0.05$ with two tails was used. Risk estimates were reported with 95% CIs.

3. Results

3.1 Study population

A total of 160,923 individuals aged 38 to 71 years who passed the initial selection criteria in the UK Biobank were included in this study (**Table 1**). The median follow-up was 7.1 (interquartile range [IQR] 6.2-8.5) years. The mean age of the included participants was 56.5 years, and 53.0% of them were women. Evidently, regular PPI users were characterized by higher rates of GERD (32.4% vs. 2.7%), peptic ulcer (5.6% vs. 0.9%), and upper gastrointestinal bleeding (0.2% vs. 0.03%) compared to non-regular PPI users. Higher burdens of comorbidities, as well as increased use of aspirin, H2RA, and cholesterol-lowering medications, were also observed in regular PPI users.

3.2 Proton pump inhibitor use and influenza, pneumonia, and COVID-19 infection

Increased risks of developing influenza, pneumonia, and other respiratory infections were identified in regular users of PPIs compared with non-regular users, and the risk remained raised after adjustments (**Table 2**). Inclusion of the self-reported cases did not significantly alter the results (**Supplementary Table S4**). A 32% increased risk of developing influenza (aHR 1.32, 95% CI 1.12-1.56, $P = 0.001$; e-value 1.97) was observed among regular PPI users. In addition, regular use of PPIs was associated with a 42% increased risk of developing pneumonia (fully adjusted HR [aHR] 1.42, 95% CI 1.26-1.59, $P < 0.001$; e-value 2.19). Regular PPI users had lower event-free probabilities for influenza and pneumonia compared to those of non-users (**Supplementary Figure 2 A-B**). The association of PPI use with COVID-19 positivity was further evaluated in our study. Initially, in the non-adjusted model, the susceptibility to COVID-19 positivity was observed with a 18% increase (HR 1.18, 95% CI 1.09-1.26, $P < 0.001$ for non-adjusted model; **Table 2**) in participants with regular use of PPIs. However, full adjustments for covariates rendered the association nonsignificant (aHR 1.08, 95% CI 0.99-1.17, $P = 0.101$; **Table 2**).

3.3 Proton pump inhibitor use and other respiratory infections, COVID-19 severity, and COVID-19 mortality

For other upper and lower respiratory infections, the risks among regular PPI users were increased by 19% (aHR 1.19, 95% CI 1.11-1.27, $P < 0.001$; e-value 1.67) and 37% (aHR 1.37, 95% CI 1.29-1.46, $P < 0.001$; e-value 2.08), respectively. In contrast, the risks of developing severe COVID-19 (aHR 1.33, 95% CI 1.09-1.61, $P = 0.004$; e value 1.99) and mortality due to COVID-19 (aHR 1.46, 95% CI 1.05-2.03, $P = 0.024$; e value 2.03) were significantly increased among PPI users compared to those among PPI non-users (**Supplementary Table S5**). PPI users had lower event-free probabilities for COVID-19 severity and mortality, but not COVID-19 positivity compared to those of non-users (**Supplementary Figure 2 C-E**).

3.4 Subgroup analysis

Stratified analyses were performed in the fully adjusted models for the main outcomes. Overall, no significant evidence of interactions was observed in the subgroup analyses referring to influenza (all P for interaction > 0.05 , **Figure 2**). The subgroup analyses for other main outcomes were illustrated in **Figure 2** and **Supplementary Figure 3**.

Among different types of PPIs, regular omeprazole or lansoprazole users were correlated with greater risks of respiratory infections (**Supplementary Table S6**). The risks of influenza were significant among *CYP2C19* normal metabolizers, and the risk estimate increased among *CYP2C19* likely intermediate, intermediate and poor metabolizers, while more information and larger sample sizes on PPI subtypes are still needed to increase the statistical power (**Supplementary Table S7**). The risks of COVID-19 severity and COVID-19 mortality were higher among *CYP2C19*

Characteristics	Regular PPI use		Overall
	Yes	No	
Number of participants, n (%)	9 997 (6.2)	150 926 (93.8)	160 923 (100.0)
Age, years, mean (SD)	59.4 (7.4)	56.3 (8.2)	56.5 (8.1)
Sex, female, n (%)	5 533 (55.4)	79 709 (52.8)	85 242 (53.0)
Ethnicity, white, n (%)	9 571 (95.7)	144 295 (95.6)	153 866 (95.6)
Deprivation index, mean (SD)	-0.9 (3.3)	-1.4 (3.0)	-1.4 (3.0)
Alcohol consumption, n (%)			
Daily or almost daily	1 805 (18.1)	28 846 (20.5)	32 874 (20.4)
3 or 4 times a week	1 923 (19.2)	33 533 (23.8)	37 570 (23.4)
1 or 2 times a week	2 396 (24.0)	37 443 (26.6)	42 228 (26.2)
1 to 3 times a month	1 179 (11.8)	15 669 (11.1)	17 988 (11.2)
Special occasions only	1 522 (15.2)	14 965 (10.6)	17 616 (11.0)
Never	1 161 (11.6)	10 414 (7.4)	12 570 (7.8)
Smoking, n (%)			
Never smoker	4 572 (45.7)	82 998 (55.0)	87 570 (54.4)
Previous smoker	4 289 (42.9)	52 013 (34.5)	56 302 (35.0)
Current smoker	1 136 (11.4)	15 915 (10.6)	17 051 (10.6)
Physical activity, MET minutes/week, median (IQR)	1 525.5 (2 722.0)	1 815.0 (2 848.5)	1 794.0 (2 838.5)
Fresh fruit intake, pieces, mean (SD)	2.0 (2.6)	1.9 (2.6)	1.9 (2.6)
BMI, kg/m ² , mean (SD)	29.2 (5.1)	27.4 (4.7)	27.5 (4.8)
Indication of PPIs, n (%)			
GERD	3 235 (32.4)	4 015 (2.7)	7 250 (4.5)
Peptic ulcer	561 (5.6)	1 303 (0.9)	1 864 (1.2)
Upper gastrointestinal bleeding	18 (0.2)	38 (0.03)	56 (0.03)
Comorbidities, n (%)			
Hypertension	4 116 (41.2)	38 162 (25.3)	42 278 (26.3)
Type 2 diabetes	124 (1.2)	890 (0.6)	1 014 (0.6)
Renal failure	60 (0.6)	243 (0.2)	303 (0.2)
Myocardial infarction	331 (3.3)	1 632 (1.1)	1 963 (1.2)
Stoke	140 (1.4)	943 (0.6)	1 083 (0.7)
COPD	46 (0.5)	200 (0.1)	246 (0.2)
Asthma	841 (8.4)	8 471 (5.6)	9 312 (5.8)
Medication use, n (%)			
Aspirin	2 457 (24.6)	21 108 (14.0)	23 565 (14.6)
Non-aspirin NSAIDs	1 224 (12.2)	22 568 (15.0)	23 792 (14.8)
H2RA	297 (3.0)	2 956 (2.0)	3 253 (2.02)
Cholesterol lowering medications	1 537 (15.4)	9 241 (6.1)	10 778 (6.70)
Multivitamin use, n (%)	2 227 (22.3)	33 201 (22.0)	35 428 (22.0)

BMI: body mass index; COPD: chronic obstructive pulmonary disease; GERD: gastroesophageal reflux disease; H2RA: histamine 2 receptor antagonist; IQR: interquartile range; MET: metabolic equivalent of task; PPI: proton pump inhibitor; NSAIDs: non-steroidal anti-inflammatory drugs; SD: standard deviation.

Table 1.

Baseline characteristics of the included participants.

	Case/person- years	Non-adjusted model		Age/sex-adjusted model		Fully adjusted model*	
		HR (95% CI)	P	HR (95% CI)	P	HR (95% CI)	P
Influenza							
Non-regular PPI use	2 009/6 011	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Regular PPI use	183/539	1.38 (1.19-1.62)	<0.001	1.49 (1.28-1.74)	<0.001	1.32 (1.12-1.56)	0.001
			1		1		
Pneumonia							
Non-regular PPI use	2 904/12 867	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Regular PPI use	378/1 702	2.04 (1.83-2.27)	<0.001	1.74 (1.56-1.94)	<0.001	1.42 (1.26-1.59)	<0.001
			1		1		
COVID-19 positivity							
Non-regular PPI use	23 989/29 080	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Regular PPI use	1 440/1 702	1.18 (1.09-1.26)	<0.001	1.07 (0.99-1.15)	0.058	1.08 (0.99-1.17)	0.101
			1				
Other upper respiratory infections							
Non-regular PPI use	14 449/52 499	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Regular PPI use	1 118/3 988	1.30 (1.22-1.38)	<0.001	1.31 (1.23-1.39)	<0.001	1.19 (1.11-1.27)	<0.001
			1		1		
Other lower respiratory infections							
Non-regular PPI use	14 494/55 384	1.00 (reference)		1.00 (reference)		1.00 (reference)	
Regular PPI use	1 486/5 598	1.78 (1.67-1.88)	<0.001	1.65 (1.56-1.74)	<0.001	1.37 (1.29-1.46)	<0.001
			1		1		

*Adjusted for age, sex, ethnicity, deprivation index, smoking, alcohol consumption, physical activity, fresh fruit intake, body mass index, any indication of PPIs (gastroesophageal reflux disease [GERD], peptic ulcer, upper gastrointestinal bleeding), comorbidities (hypertension, type 2 diabetes, renal failure, myocardial infarction, stroke, chronic obstructive pulmonary disease [COPD], asthma), medications (aspirin, non-aspirin non-steroidal anti-inflammatory drugs [NSAIDs, ibuprofen], histamine 2 receptor antagonists (H2RAs), cholesterol lowering medications), multivitamin use, and influenza vaccination (for influenza) or COVID-19 vaccination (for COVID-19-related outcomes).

CI: confidence interval; COVID-19: coronavirus disease 2019; HR: hazard ratio; PPI: proton pump inhibitor.

Table 2.

Associations of PPI use with the susceptibility to pneumonia, influenza, COVID-19 positivity, and other respiratory infections.

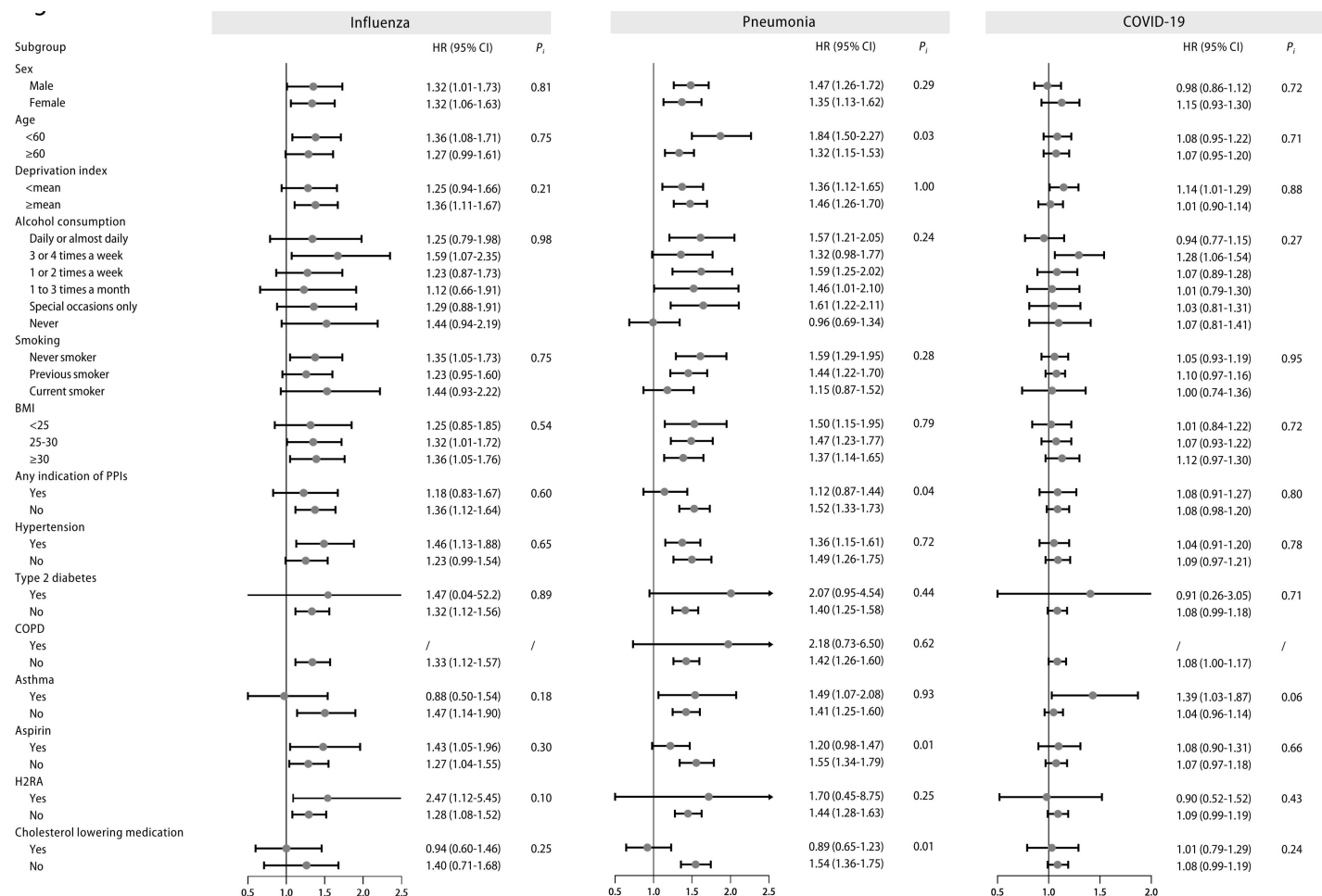


Figure 2.

Stratified analysis of regular proton pump inhibitor (PPI) users and the risk of influenza, pneumonia, and COVID-19 infection.

Effect estimates were based on age, sex, deprivation index, alcohol consumption, smoking, body mass index (BMI), indications of PPIs, hypertension, type 2 diabetes, chronic obstructive pulmonary disease (COPD), asthma, aspirin, histamine 2 receptor antagonist (H2RA), and cholesterol-lowering medication, using the fully adjusted model. CI: confidence interval; HR: hazard ratio; P_i : P value for interaction.

likely intermediate, intermediate and poor metabolizers (**Supplementary Table S8**). The risks of pneumonia were higher among *CYP2C19* rapid and ultrarapid metabolizers (**Supplementary Table S7**).

3.5 Analysis by multiple imputation and propensity score-matching

After imputation of missing data, we found that individuals with regular use of PPIs were associated with similarly increased trends in the risks of influenza, pneumonia, other upper respiratory infections, and other lower respiratory infections (all $P < 0.05$) (**Supplementary Table S9**). The associations with COVID-19 severity and mortality were also significant (all $P < 0.05$) (**Supplementary Table S10**).

Matching of 9,910 regular PPI users and 39,760 non-regular users (1:4 by propensity scores) was also conducted, and the baseline characteristics were much more similar (**Supplementary Table S11**). The participants regularly exposed to PPIs were observed with increased risks for influenza, pneumonia, other upper respiratory infections, and other lower respiratory infections (all $P < 0.05$) (**Supplementary Table S12**), which were consistent with the results from Cox hazard proportional regression models. The associations with COVID-19 severity and mortality were also significant (all $P < 0.05$) (**Supplementary Table S13**).

3.6 Comparisons with H2RA users

To further confirm the results and reduce the effect of confounding by indications, we evaluated the risk of respiratory infections compared to the use of H2RAs, which is a less potent acid-suppressant and contains indications similar to PPI. When compared to regular H2RA users, participants with regular use of PPIs were also associated with an increased risk of influenza (HR 1.74, 95% CI 1.19-2.54, $P = 0.004$; e-value 2.87), other upper respiratory infection (HR 1.28, 95% CI 1.07-1.54, $P = 0.008$; e-value 1.88), and other lower respiratory infection (HR 1.33, 95% CI 1.18-1.50, $P < 0.001$; e-value 1.99) (**Table 3**). However, the associations with pneumonia (HR 1.22, 95% CI 0.96-1.54, $P = 0.104$), COVID-19 infection (HR 1.04, 95% CI 0.87-1.26, $P = 0.629$), COVID-19 severity (HR 0.91, 95% CI 0.64-1.30, $P = 0.608$), or COVID-19 mortality (HR 0.83, 95% CI 0.45-1.56, $P = 0.745$) were not significant (**Supplementary Table S14**).

4. Discussion

In this large-scale, population-based cohort with updated information, we identify that the use of PPIs is associated with incident influenza. In contrast, analyses of pneumonia, COVID-19 infection and related outcomes, reveal attenuated effects after being compared with H2RA users. The association with influenza remains robust across different subgroups stratified by population characteristics and *CYP2C19* phenotypes.

The correlation between PPI use and the risk of influenza remains unexplored. For the past two decades, accumulating evidence indicates increased risks of pneumonia under the use of PPIs [7, 8, 28, 29], whereas others failed to show such associations [9, 10]. Conflicting findings also exist for studies referring to the risk of COVID-19 infection and related outcomes among PPI users, including several meta-analyses [11–15, 30–33]. Compared with existing studies, our study more comprehensively adjusts for a variety of critical covariates by utilizing the latest data from the UK Biobank. In addition, distinct from previous population-based studies, we compared the risks with H2RA users to further reduce protopathic and other unmeasured bias, since the users of acid suppressants, including PPIs and H2RAs, can have matched information on different characteristics, including indications. Although the risks of

	Cases / Person-years	HR (95% CI)*	P
Influenza			
Regular H2RA use	32/102	1.00 (Reference)	
Regular PPI use	175/524	1.74 (1.19-2.54)	0.004
Pneumonia			
Regular H2RA use	86/385	1.00 (Reference)	
Regular PPI use	368/1653	1.22 (0.96-1.54)	0.104
COVID-19 positivity			
Regular H2RA use	425/506	1.00 (Reference)	
Regular PPI use	1 409/1 665	1.06 (0.90-1.24)	0.509
Other upper respiratory infection			
Regular H2RA use	146/522	1.00 (Reference)	
Regular PPI use	602/2099	1.28 (1.07-1.54)	0.008
Other lower respiratory infection			
Regular H2RA use	339/1350	1.00 (Reference)	
Regular PPI use	1438/5398	1.33 (1.18-1.50)	<0.001

CI: confidence interval; COVID-19: coronavirus disease 2019; H2RA: histamine-2 receptor antagonist; HR: hazard ratio; PPI: proton pump inhibitor.

*Adjusted for age, sex, ethnicity, deprivation index, smoking, alcohol consumption, physical activity, fresh fruit intake, body mass index, any indication of PPIs (gastroesophageal reflux disease [GERD], peptic ulcer, upper gastrointestinal bleeding), comorbidities (hypertension, type 2 diabetes, renal failure, myocardial infarction, stroke, chronic obstructive pulmonary disease [COPD], asthma), medications (aspirin, non-aspirin non-steroidal anti-inflammatory drugs [NSAIDs, ibuprofen], cholesterol lowering medications), multivitamin use, and influenza vaccination (for influenza) or COVID-19 vaccination (for COVID-19-related outcomes).

Table 3.

Comparisons of the risks of influenza, pneumonia, and COVID-19 between proton pump inhibitor (PPI) and histamine-2 receptor antagonist (H2RA) users.

pneumonia were initially increased in Cox and propensity-score-matched analyses, direct comparison with H2RA users showed negative results, which indicates that previously observed associations could be due to unmeasured confounders.

Several proposed mechanisms can account for the association between the use of PPIs and respiratory tract infections. Since a low pH of gastric acid rapidly inactivates microorganisms, one critical issue is that reduced acidity induced by PPIs leads to the overgrowth of microorganisms, which can contribute to the development of infections in the respiratory tract through microaspiration[34]. Colonization and growth of pathogens under hypochlorhydria could increase the risk of respiratory infections. Although initial assessments indicated the use of PPIs might increase the risk of pneumonia, the head-to-head comparison with H2RAs yielded impacted effects. It could be due to the similar acid-suppressive effects of H2RAs and reduced sample size, which therefore warrants further investigations.

Concerns over protopathic bias due to non-specific and overlapping symptoms between influenza/pneumonia and acid-related diseases were raised [35]. Nevertheless, pneumonia and influenza often present with acute cough, and other concomitant symptoms distinct from acid-related diseases [36]. In contrast, patients with chronic cough are more commonly GERD-related [36]. The American College of Chest Physicians Clinical Practice Guidelines for Management of Reflux-Cough Syndrome also recommend against using PPI therapy alone for patients with chronic cough but without heartburn or regurgitation [37]. In our study, the use of PPIs is defined as taking the medication for most days of the week in the last 4 weeks, which is uncommon for acute cough. Although we cannot completely rule out protopathic bias, we have attempted to minimize it by adjusting for covariates including PPI indications, matching with propensity scores, and comparing with H2RA users.

For the risk of developing influenza, we analyzed the risks among different *CYP2C19* metabolizers for the first time, and further observed a significant increase among *CYP2C19* normal metabolizers compared to rapid and ultrarapid metabolizers. Although the risks of several outcomes, for example, influenza and pneumonia, for *CYP2C19* likely intermediate, intermediate and poor metabolizers are not statistically significant, they could be due to the limited sample size, and the risk estimates are higher compared to those among other types of metabolizers. Intriguingly, the risks of developing influenza and pneumonia are higher among *CYP2C19* rapid and ultrarapid metabolizers regularly taking PPIs compared to other types of metabolizers. Since our study exclusively involves participants with valid primary care data, such an increased risk might be to some extent contributed by the over-prescription or self-taking of PPIs under the undesired effects without following the proper strategy. Our findings are generally consistent with the assumption that slower metabolizers are associated with higher risks of adverse effects, while larger samples are needed to increase statistical power. Prescription of PPIs based on different *CYP2C19* metabolism subtypes is therefore important to reduce the adverse effects.

Our study has several strengths. First, our study utilizes the updated large-sample data from the UK Biobank and exclusively includes participants with valid records from primary care, which reduces the information bias. Second, a variety of covariates, especially for the indications of PPIs and the use of aspirin, which might contribute to indication or protopathic bias, have been adjusted to enhance the robustness of our results. Third, genotypic data of metabolic enzymes has been integrated into our study. Fourth, propensity score-matching analysis reduces a greater portion of bias, and analyses by propensity-score matching or multiple imputation derive consistent results. Fifth, adjustments for vaccination for COVID-19 and influenza has been performed in our study to reduce the confounding effects by vaccination. Furthermore, the comparison with participants using H2RA, a less potent acid suppressant with similar indications, further reduces the confounding by indication. The findings on the risk of influenza remain highly consistent across different strata and sensitivity analyses.

Limitations exist in our study. Information on dose and duration of PPI use, discrimination between prescription and over-the-counter use of PPIs, health-seeking behavior, different types of pneumonia, and pneumococcus vaccination is currently not available from the UK Biobank. Given that the PPI exposure was mainly assessed at the baseline recruitment, it was possible that a small proportion of PPI users was misclassified during the follow-up due to the medication discontinuation, which may result in an underestimation of potential risk. However, the prevalent user design could underestimate the actual risks of PPI use for respiratory infections, which indicates the real effect might be stronger [38]. In addition, no effect moderation was observed in subgroup analyses for the main outcome among PPI users with indications (more likely to regularly use PPIs for a long period) compared to those without indications, indicating the risks remained increased among long-term PPI users. Since the follow-up prescription data was lacking in our study to precisely identifying the long-term users, further evaluation using cohorts with close follow-up is needed. PPIs are indicated for *Helicobacter pylori* (*H. pylori*) eradication, whereas the UK Biobank does not contain adequate data. Thus, the indication for eradicating *H. pylori* is not adjusted in this study. The data on different PPI subtypes and COVID-19 infection and related outcomes are relatively small, which limits their power and still needs further investigation. Moreover, patients with exacerbations of comorbid disorders (e.g., diabetes, asthma, COPD) might suffer from a wide range of gastrointestinal symptoms that lead to the use of PPIs [39] (Supplementary Figure S4). Due to the lack of data for respiratory severity and close follow-up for medication use, residual confounding might still exist due to the observational nature. Residual genotyping impacts of other enzymes, although affecting the metabolism to a lesser extent, might also exist. Although no significant differences were found between PPIs and H2RAs regarding the association with pneumonia and COVID-19-related outcomes, this could be due to the reduced sample size and power, which require larger cohorts to validate the effects.

Furthermore, the highly selective nature of the UK Biobank might create collider stratification bias for the evaluation of COVID-19-related outcomes, and thus the conclusions should be interpreted with cautions [40]. The current study is principally based on white British ancestry in the United Kingdom, and future exploration of other ancestries with comparisons is warranted.

Our findings could have essential implications for the prevention of respiratory infections and the de-prescribing of PPIs in clinical practice. Administration of PPIs can rapidly increase intragastric pH to higher than 6 after 2-4 hours [41]. Emerging evidence has revealed the inappropriate prescription of PPIs in both the primary and secondary care settings, and 33-67% of the patients did not take the drug according to their countries' criteria [42]. Similarly, the baseline characteristics of the included participants in our study demonstrate that approximately 60% of the regular PPI users do not have main indications. In addition, although influenza is usually self-limiting in healthy individuals, its risk of complications is significantly increased among pregnant women and people with immunosuppression or chronic diseases [43]. Therefore, comprehensive evaluation of PPI use is needed in clinical practice.

5. Conclusion

In conclusion, compared to non-users, people regularly taking PPIs are associated with increased susceptibility to influenza, pneumonia, as well as COVID-19 severity and mortality, while their association with pneumonia and COVID-19-related outcomes is diminished after comparison with H2RA use and remains to be further explored.

Conflict of Interest Statement

The authors declared no conflict of interest.

Declaration of generative AI in scientific writing

This study does not involve generative AI in scientific writing.

Author Contributions

RJZ, YYM, LJZ, and DLL contributed equally to this work. HC, WHS, RJZ, YYM, and LJZ contributed to data extraction, data analyses, and manuscript drafting. DLL contributed to data interpretation and manuscript drafting. HHW, ZWZ, QY, and JWL contributed to manuscript drafting. HC, WHS, QC, and FWL contributed to study design, data interpretation, and final approval of the manuscript.

Data Availability

The UK Biobank data are available at <https://www.ukbiobank.ac.uk/> after approval of application.

Acknowledgements

The authors are grateful to the UK Biobank for approval and access to data of the project, and this research has been conducted under Application Number 83339. We thank Dr. Qian Chen for her kind suggestions on statistical analyses.

Availability of Data and Materials

The UK Biobank data are available on application to the UK Biobank (www.ukbiobank.ac.uk). The codes used in this study can be found at: <https://epirhandbook.com/en/> and https://cran.r-project.org/doc/contrib/Epicalc_Book.pdf.

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Abbreviations

- ACE2: angiotensin-converting enzyme 2
- BMI: body mass index
- CI: confidence interval
- COVID-19: Coronavirus Disease 2019
- DCA: directed acyclic graph
- GERD: gastroesophageal reflux disease
- H2RA: histamine-2 receptor antagonist
- HR: hazard ratio
- IQR: interquartile range
- NSAIDs: non-aspirin non-steroidal anti-inflammatory drugs
- PPI: proton pump inhibitor
- SARS-CoV-2: severe acute respiratory syndrome coronavirus 2
- SD: standard deviation.

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Editors

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Reviewer #1 (Public Review):

Summary:

The current study aims to quantify associations between regular use of proton-pump inhibitors (PPI) - defined as using PPI most days of the week during the last 4 weeks at one

cross-section in time - with several respiratory outcomes (6 in total: risk of influenza, pneumonia, COVID-19, other respiratory tract infections, as well as COVID-19 severity and mortality) up to several years later in time.

Strengths:

Several sensitivity analyses were performed, including i) estimation of the e-value to assess how strong unmeasured confounders should be to explain observed effects, ii) comparison with another drug with a similar indication to potentially reduce (but not eliminate) confounding by indication, iii)

Weaknesses:

While the original submission had several weaknesses, the authors have appropriately addressed all issues raised. There are inevitable weaknesses remaining, but these are appropriately highlighted in the discussion. Remaining weaknesses that remain - but are highlighted in the discussion - include the fact that the main exposure of interest is only measured at one time-point whereas outcomes are assessed over a long time period, the inclusion of prevalent users leading to potential bias (e.g. those experiencing bad outcomes already stopping because of side-effects before inclusion in the study), and the possibility of unmeasured confounding explaining observations (e.g. severity of underlying comorbidities leading to PPI prescriptions combined with the absence of information about comorbidity severity), and potential selection bias.

<https://doi.org/10.7554/eLife.94973.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

The current study aims to quantify associations between the regular use of proton-pump inhibitors (PPI) - defined as using PPI most days of the week during the last 4 weeks at one cross-section in time - with several respiratory outcomes up to several years later in time. There are 6 respiratory outcomes included: risk of influenza, pneumonia, COVID-19, other respiratory tract infections, as well as COVID-19 severity and mortality).

Strengths:

Several sensitivity analyses were performed, including i) estimation of the e-value to assess how strong unmeasured confounders should be to explain observed effects, ii) comparison with another drug with a similar indication to potentially reduce (but not eliminate) confounding by indication.

We are grateful for your pointing out the strengths in our article, particularly the assessment of e-values and the comparison with another medication to mitigate confounding by indication. We extend our sincere gratitude to the reviewer for identifying multiple concerns and offering constructive feedback to help improve our manuscript. We will incorporate these suggestions into our revisions.

Weaknesses:

(1) The main exposure of interest seems to be only measured at one time-point in time (at study enrollment) while patients are considered many years at risk afterwards without knowing their exposure status at the time of experiencing the outcome. As indicated by the authors, PPI are sometimes used for only short amounts of time. It seems biologically implausible that an infection was caused by using PPI for a few weeks many years ago.

We agree with the reviewer that PPIs are sometimes used for only short amounts of time, as indicated in our manuscript. We acknowledge that it is a limitation of the UK Biobank cohort, and we have discussed this in the discussion section as follows:

“Given that the PPI exposure was mainly assessed at the baseline recruitment, it was possible that a small proportion of PPI users was misclassified during the follow-up due to the medication discontinuation, which may result in an underestimation of potential risk.” (Page 14, Line 8-10)

In addition, to alleviate these concerns, we have conducted effect medication for the subgroup of potential long-term users, which were defined by participants with indications of PPI use. This information has been included in the discussion section:

“In addition, no effect moderation was observed in subgroup analyses for the main outcome among PPI users with indications (more likely to regularly use PPIs for a long period) compared to those without indications, indicating the risks remained increased among long-term PPI users.” (Page 14, Line 12-15)

We hope that in the future, the concerns highlighted by the reviewer can be resolved by utilizing datasets with close follow-up, especially regarding medication use:

“Since the follow-up prescription data was lacking in our study to precisely identifying the long-term users, further evaluation using cohorts with close follow-up is needed.” (Page 14, Line 15-17)

(2) Previous studies have shown that by focusing on prevalent users of drugs, one often induces several biases such as collider stratification bias, selection bias through depletion of susceptible, etc.

Because of the limitations of data from the UK Biobank, such as the absence of details on initiation of medications and regular monitoring, we were restricted to using a prevalent user design to assess the associations between PPI use and respiratory outcomes. We have discussed it in the limitation section:

“Given that the PPI exposure was mainly assessed at the baseline recruitment, it was possible that a small proportion of PPI users was misclassified during the follow-up due to the medication discontinuation, which may result in an underestimation of potential risk. However, the prevalent user design could underestimate the actual risks of PPI use for respiratory infections, which indicates the real effect might be stronger [38].....Since the follow-up prescription data was lacking in our study to precisely identifying the long-term users, further evaluation using cohorts with close follow-up is needed.” (Page 14, Line 8-17)

(3) It seems Kaplan Meier curves are not adjusted for confounding through e.g. inverse probability weighting. As such the KM curves are currently not informative (or the authors need to make clearer that curves are actually adjusted for measured confounding).

Your kind suggestions are greatly appreciated. We have plotted Kaplan Meier curves adjusted for confounding by inverse probability weighting with the measured confounders according

to the reviewer's advice. The methods and results are demonstrated as follows:

"The event-free probabilities were compared by Kaplan-Meier survival curves with inverse probability weights adjusting for the measured covariates." (Page 8, Line 13-15)

"Regular PPI users had lower event-free probabilities for influenza and pneumonia compared to those of non-users (Supplementary Figure 2 A-B)." (Page 9, Line 21-23)

"PPI users had lower event-free probabilities for COVID-19 severity and mortality, but not COVID-19 positivity compared to those of non-users (Supplementary Figure 2 C-E)." (Page 10, Line 9-10)

(4) Throughout the manuscript the authors seem to misuse the term multivariate (using one model with e.g. correlated error terms to assess multiple outcomes at once) when they seem to mean multivariable.

We apologize for misusing the term "multivariate" and "multivariable" in our previous manuscript. We have corrected the misused terms throughout the manuscript:

"Univariate and multivariable Cox proportional hazards regression models were utilized to assess the association between regular use of PPIs and the selected outcomes." (Page 7, Line 19-20)

"The remaining imbalanced covariates (standardized mean difference ≥ 0.1) after propensity score matching were further adjusted by multivariate multivariable Cox regression models to calculate HRs and 95% CIs." (Page 8, Line 23-25)

(5) Given multiple outcomes are assessed there is a clear argument for accounting for multiple testing, which following the logic of the authors used in terms of claiming there is no association when results are not significant may change their conclusions. More high-level, the authors should avoid the pitfall of stating there is evidence of absence if there is only an absence of evidence in a better way (no statistically significant association doesn't mean no relationship exists).

We have revised our interpretation for the results, particularly for those without statically significant association based on the reviewer's advice, and clearly recognize that the conclusions should be interpreted with cautions:

"In contrast, the risk of COVID-19 infection was not significant with regular PPI use..." (Page 2, Line 11-12)

"PPI users were associated with a higher risk of influenza (HR 1.74, 95%CI 1.19-2.54), but the risks with pneumonia or COVID-19-related outcomes were not evident." (Page 2, Line 14-16)

"...while the effects on pneumonia or COVID-19-related outcomes under PPI use were attenuated when compared to the use of H2RAs." (Page 2, Line 18-19, in the Abstract)

"...while their association with pneumonia and COVID-19-related outcomes is diminished after comparison with H2RA use and remains to be further explored." (Page 15, Line 21-22, in the Conclusion)

(6) While the authors claim that the quantitative bias analysis does show results are robust to unmeasured confounding, I would disagree with this. The e-values are around 2 and it is clearly not implausible that there are one or more unmeasured risk factors that together or alone would have such an effect size. Furthermore, if one would use the same (significance) criteria as used by the authors for determining whether an

association exists, the required effect size for an unmeasured confounder to render effects 'statistically non-significant' would be even smaller.

We agree with the reviewer that there might still exist one or more unmeasured risk factors that have effect sizes larger than 2. Hence, we cannot affirm that the findings are robust to unmeasured confounding in the current analysis, which is a limitation of our study. We have deleted the previous statement, and added more discussion in the limitation section:

“Moreover, patients with exacerbations of respiratory disorders (e.g., asthma, COPD) might suffer from a wide range of gastrointestinal symptoms that lead to the use of PPIs [38]. Due to the lack of data for respiratory severity and close follow-up for medication use, residual confounding might still exist due to the observational nature.” (Page 14, Line 23-27)

(7) Some patients are excluded due to the absence of follow-up, but it is unclear how that is determined. Is there potentially some selection bias underlying this where those who are less healthy stop participating in the UK biobank?

Thank you for your question. The reasons for the absence of follow-up are mainly classified into five categories, including: (1) Death reported to UK Biobank by a relative; (2) NHS records indicate they are lost to follow-up; (3) NHS records indicate they have left the UK; (4) UK Biobank sources report they have left the UK; (5) Participant has withdrawn consent for future linkage. According to the data from UK Biobank (<https://biobank.ndph.ox.ac.uk/showcase/field.cgi?id=190>), the major reason for the loss of follow-up among participants is their departure from the UK (84.7% of participants who were lost to follow-up). In addition, not including those who were less healthy in the study might also underestimate the risk, leading to lower estimated effects of PPIs for respiratory infections. We have supplemented this in our revised manuscript:

“Among them, 1,297 participants without follow-up, which were mainly determined by reported death, departure from the UK, or withdrawn consent, had been removed after initial exclusion.” (Page 4, Line 25-27)

(8) Given that the exposure is based on self-report how certain can we be that patients e.g. do know that their branded over-the-counter drugs are PPI (e.g. guardium tablets)? Some discussion around this potential issue is lacking.

Thank you for your concerns. In the data collection by the UK Biobank, the participants can enter the generic or trade name of the treatment on the touchscreen to match the medications they used. We have added this important information to the method section:

“The exposure of interest was regular use of PPIs. The participants could enter the generic or trade name of the treatment on the touchscreen to match the medications they used (Supplementary Table S1).” (Page 5, Line 6-8)

We acknowledge that specific information on prescribed or over-the-counter use of medications is lacking in the UK Biobank. We have discussed it in the limitation section:

“Limitations exist in our study. Information on dose and duration of PPI use, discrimination between prescription and over-the-counter use of PPIs, health-seeking behavior, different types of pneumonia, and pneumococcus vaccination is currently not available from the UK Biobank.” (Page 14, Line 5-8)

(9) Details about the deprivation index are needed in the main text as this is a UK-specific variable that will be unfamiliar to most readers.

Thank you for your question on the definition of deprivation index. We have proved the details about the deprivation index in the manuscript:

“...socioeconomic status (deprivation index, which was defined using national census information on car ownership, household overcrowding, owner occupation, and unemployment combined for postcode areas of residence)...” (Page 6, Line 14-17)

(10) It is unclear how variables were coded/incorporated from the main text. More details are required, e.g. was age included as a continuous variable and if so was non-linearity considered and how?

We apologize for not elucidating how variables were incorporated into the main text. Previously, the linearity between continuous variables and outcomes was assessed by Martingale residuals plots, while the variables detected with non-linearity were regarded as categorical variables for further analyses. For example, after evaluation with the Martingale residuals plot, age demonstrated non-linearity, and we incorporated it as a categorical variable for the analysis of COVID-related mortality.

We have supplemented the information in the method section:

“The linearity between continuous variables and outcomes was assessed by Martingale residuals plots, while the variables detected with non-linearity were regarded as categorical variables for further analyses.” (Page 6, Line 28 to Page 7, Line 1)

(11) The authors state that Schoenfeld residuals were tested, but don't report the test statistics. Could they please provide these, e.g. it would already be informative if they report that all p-values are above a certain value.

We are sorry for not providing the statistics about the Schoenfeld residual in our previous manuscript. We have supplemented the information in our revisions:

“Schoenfeld residuals tests were used to evaluate the proportional hazards assumptions, while no violation of the assumption was detected (Supplementary Table S3).” (Page 7, Line 27 to Page 8, Line 1)

(12) The authors would ideally extend their discussion around unmeasured confounding, e.g. using the DAGs provided in <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7832226/>, in particular (but not limited to) around severity and not just presence/absence of comorbidities.

Thank you for your insightful suggestions that the discussion about unmeasured confounding should be extended. We agree with the reviewer that, in addition to the comorbidities themselves, their severity could also have an important impact on the use of PPIs. We have added the discussion in the limitation section with citing the article (PMC7832226):

“Moreover, patients with exacerbations of comorbid disorders (e.g., diabetes, asthma, COPD) might suffer from a wide range of gastrointestinal symptoms that lead to the use of PPIs [38] (Supplementary Figure S4). Due to the lack of data for respiratory severity and close follow-up for medication use, residual confounding might still exist due to the observational nature.” (Page 14, Line 23-27)

(13) The UK biobank is known to be highly selected for a range of genetic, behavioural, cardiovascular, demographic, and anthropometric traits. The potential problems this might create in terms of collider stratification bias - as highlighted here for example:

<https://www.nature.com/articles/s41467-020-19478-2> - should be discussed in greater detail and also appreciated more when providing conclusions.

We acknowledge the reviewer's point about the UK Biobank's highly selective nature potentially leading to collider stratification bias in the evaluation of COVID-19-related outcomes. We have discussed this in detail and are cautious when generating conclusions.

“Furthermore, the highly selective nature of the UK Biobank might create collider stratification bias for the evaluation of COVID-19-related outcomes, and thus the conclusions should be interpreted with cautions [39].” (Page 15, Line 2-4)

Reviewer #2 (Public Review):

Summary:

Zeng et al investigate in an observational population-based cohort study whether the use of proton pump inhibitors (PPIs) is associated with an increased risk of several respiratory infections among which are influenza, pneumonia, and COVID-19. They conclude that compared to non-users, people regularly taking PPIs have increased susceptibility to influenza, pneumonia, as well as COVID-19 severity and mortality. By performing several different statistical analyses, they try to reduce bias as much as possible, to end up with robust estimates of the association.

Strengths:

The study comprehensively adjusts for a variety of critical covariates and by using different statistical analyses, including propensity-score-matched analyses and quantitative bias analysis, the estimates of the associations can be considered robust.

We are grateful to the reviewer for pointing out the merits of our articles, which include adjusting for a wide range of covariates, employing diverse statistical analyses, and using robust data. We will revise our manuscript further based on the reviewer's suggestions.

Weaknesses:

As it is an observational cohort study there still might be bias. Information on the dose or duration of acid suppressant use was not available, but might be of influence on the results. The outcome of interest was obtained from primary care data, suggesting that only infections as diagnosed by a physician are taken into account. Due to the self-limiting nature of the outcome, differences in health-seeking behavior might affect the results.

Thank you for your questions for information on the dose/duration of acid suppressants, the source of diagnosis, and the health-seeking behavior of participants. For the data from the UK Biobank, the dose or duration of acid suppressant use was not available since the information was not collected as baseline or follow-up. In addition, the outcome of interest was also retrieved from the hospital ICD diagnosis. We apologize for not clarifying it in our previous manuscript. Moreover, we agree with the reviewer that the health-seeking behavior could have an impact on the analyses, whereas the correlated data are still not available from the UK Biobank. We have discussed them in the method and limitation section:

“Briefly, the first reported occurrences of respiratory system-related conditions within primary care data, and hospital inpatient data defined by the International Classification of Diseases (ICD)- 10 codes were categorized by the UK Biobank.” (Page 5, Line 21-25)

“Limitations exist in our study. Information on dose and duration of PPI use, discrimination between prescription and over-the-counter use of PPIs, health-seeking behavior, different

types of pneumonia, and pneumococcus vaccination is currently not available from the UK Biobank.” (Page 14, Line 5-8)

Reviewer #1 (Recommendations For The Authors):

Analysis code should be made available.

Thank you for your question. We have provide the sources of the analysis code we used for this study in our revised manuscript:

“The codes used in this study can be found at: <https://epirhandbook.com/en/> and https://cran.r-project.org/doc/contrib/Epicalc_Book.pdf.” (Page 16, Line 21-22)

Reviewer #2 (Recommendations For The Authors):

It might be interesting to study whether including self-reported infections changes the results, as people using PPI may more easily consult their GP even for a self-limiting disease such as influenza and therefore are more likely diagnosed/confirmed with such a respiratory infection.

Thank you for your insightful suggestions on conducting analyses including self-reported infections. Therefore, we have included the self-reported cases as sensitivity analyses, and the results were not significantly altered, which confirms the robustness of our results:

“Self-reported infections, except for COVID-19-related outcomes due to the lack of data, were also included for the outcomes as sensitivity analyses. The self-reported cases were reported at the baseline or subsequent UK Biobank assessment center visit.” (Page 8, Line 17-19)

“Inclusion of the self-reported cases did not significantly alter the results (Supplementary Table S4).” (Page 9, Line 17-18)

Moreover, to address the above-mentioned, sub-analyses differentiating between over-the-counter and prescribed medication might be interesting.

Thank you for your questions on differentiating between over-the-counter and prescribed medication. We have thoroughly looked up the data provided by the UK Biobank, but it is a pity that they are not provided. We have discussed this in the limitation section:

“Information on dose and duration of PPI use, discrimination between prescription and over-the-counter use of PPIs, health-seeking behavior, different types of pneumonia, and pneumococcus vaccination is currently not available from the UK Biobank.” (Page 14, Line 5-8)

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